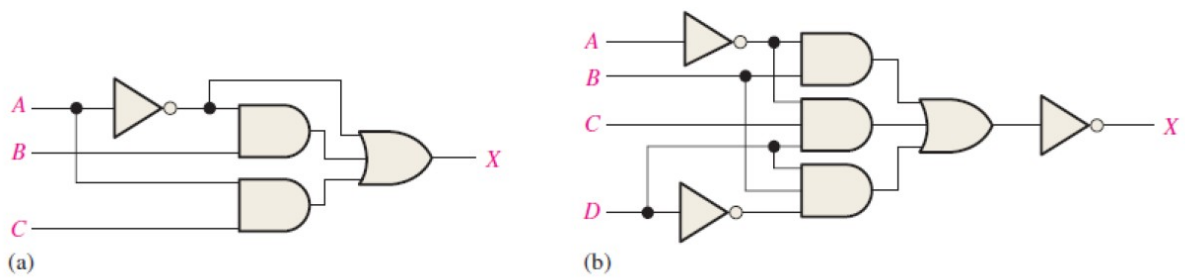
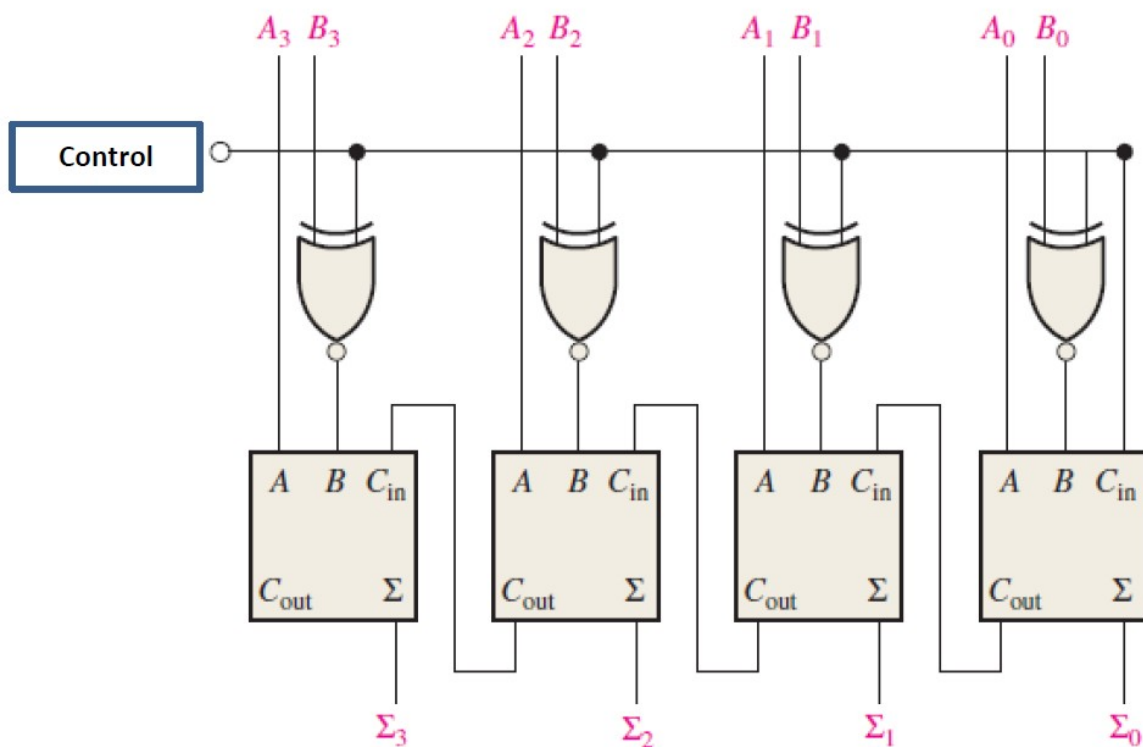


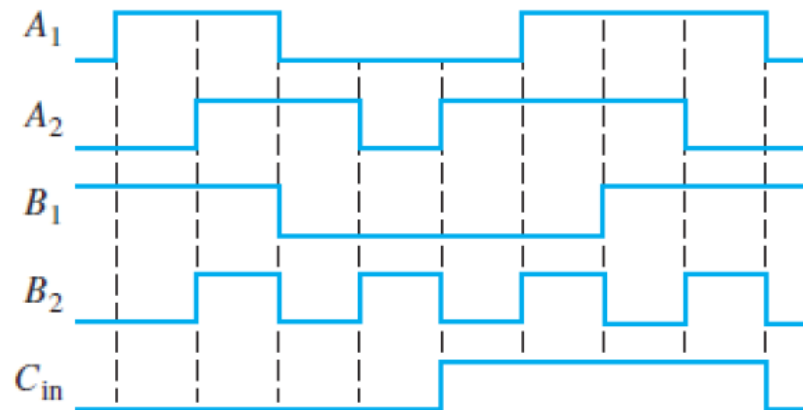
1. Write output expressions.



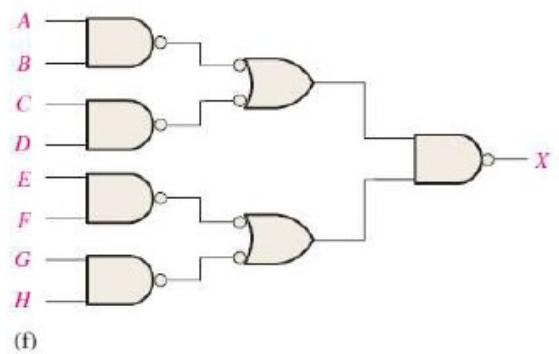
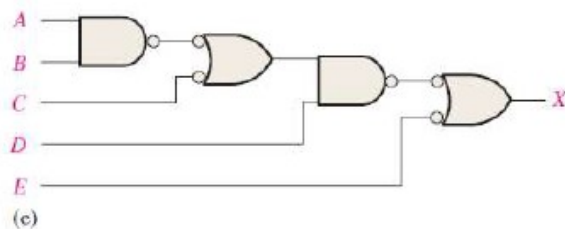
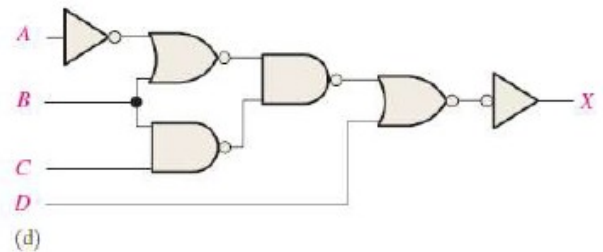
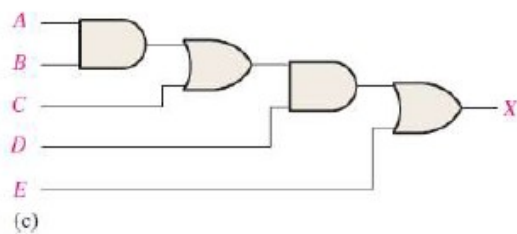
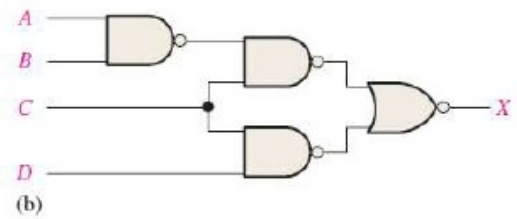
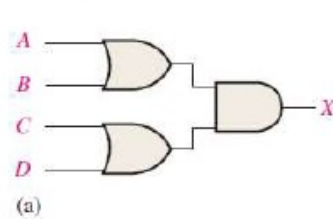
2. The circuit shown is a 4-bit circuit that can add or subtract numbers in a form used in computers (positive numbers in true form; negative numbers in complement form). (a) Explain what happens when the Control input is HIGH. (b) What happens when Control is LOW? Write equations for both situations.



3. The inputs below are applied to a 2-bit adder. Draw the output waveform/s



4. Write the output expression for each circuit and then change each circuit to an equivalent (a) AND-OR configuration (b) minimum number of gates



5. Implement logic circuit for following truth tables:

Table 1

Inputs				Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>X</i>
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Table 2

Inputs			Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

6. Draw the output waveform



Figure 1

7. Identify which Boolean rule is used in each of the following equalities

(a) $\overline{\overline{AB} + \overline{CD} + \overline{EF}} = AB + CD + EF$

(b) $A\overline{A}B + A\overline{B}\overline{C} + AB\overline{B} = A\overline{B}\overline{C}$

(c) $A(BC + \overline{BC}) + AC = A(BC) + AC$

(d) $AB(C + \overline{C}) + AC = AB + AC$

(e) $A\overline{B} + A\overline{B}C = A\overline{B}$

(f) $ABC + \overline{A}\overline{B} + \overline{A}\overline{B}\overline{C}D = ABC + \overline{A}\overline{B} + D$

8. Apply DeMorgan's theorem to the following and simplify:

1	$\overline{AB(C + D)}$
2	$\overline{(A + \overline{B} + C + \overline{D}) + ABC\overline{D}}$
3	$\overline{AB(CD + \overline{EF})(\overline{AB} + \overline{CD})}$
4	$\overline{AB(CD + EF)}$
5	$\overline{(\overline{A} + B + C + D)(\overline{ABCD})}$
6	$\overline{(\overline{ABC})(\overline{EFG}) + (\overline{HIJ})(\overline{KLM})}$
7	$\overline{(A + \overline{BC} + CD) + \overline{BC}}$
8	$\overline{(\overline{A} + B)(\overline{C} + D)(\overline{E} + F)(\overline{G} + H)}$
9	$\overline{A(B + C)}$

9. Simplify the following expressions and mention which Boolean theorem is used in each step:

Set 1:

- (a) $A(A + B)$ (b) $A(\overline{A} + AB)$ (c) $BC + \overline{BC}$
 (d) $A(A + \overline{AB})$ (e) $\overline{ABC} + \overline{ABC} + \overline{A}\overline{BC}$

Set 2:

- (a) $(\overline{A} + B)(A + C)$ (b) $\overline{AB} + \overline{ABC} + \overline{ABCD} + \overline{ABCDE}$
 (c) $BC + \overline{BCD} + B$ (d) $(B + \overline{B})(BC + B\overline{C})$
 (e) $BC + (\overline{B} + \overline{C})D + BC$

Set 3:

- (a) $CE + C(E + F) + \overline{E}(E + G)$ (b) $\overline{BCD} + (\overline{B} + \overline{C} + \overline{D}) + \overline{BCDE}$
 (c) $(C + CD)(C + \overline{CD})(C + E)$ (d) $BCDE + BC(\overline{DE}) + (\overline{BC})DE$
 (e) $BCD[BC + \overline{D}(CD + BD)]$

10. Derive standard SOP and POS expressions for following Truth tables:

ABC		ABC		$ABCD$	X	$ABCD$	X
000	0	000	0	0000	1	0000	0
001	1	001	0	0001	1	0001	0
010	0	010	0	0010	0	0010	1
011	0	011	0	0011	1	0011	0
100	1	100	0	0100	0	0100	1
101	1	101	1	0101	1	0101	1
110	0	110	1	0110	1	0110	0
111	1	111	1	0111	0	0111	1
				1000	0	1000	0
				1001	1	1001	0
				1010	0	1010	0
				1011	0	1011	1
				1100	1	1100	1
				1101	0	1101	0
				1110	0	1110	0
				1111	0	1111	1
(a)		(b)		(c)		(d)	

11. Using K-Maps, simplify the following Boolean function F , together with the don't-care conditions d , and then express the simplified function in sum-of-minterms and product-of-maxterms form:

(a) $F(x, y, z) = \Sigma(0, 1, 3, 5, 7)$

$d(x, y, z) = \Sigma(2, 4, 6)$

(c) $F(A, B, C, D) = \Sigma(5, 6, 7, 11, 14, 15)$

$d(A, B, C, D) = \Sigma(3, 9, 13)$

(b)* $F(A, B, C, D) = \Sigma(0, 4, 8, 10, 14)$

$d(A, B, C, D) = \Sigma(2, 6, 12)$

(d) $F(A, B, C, D) = \Sigma(2, 4, 7, 10, 12, 14)$

$d(A, B, C, D) = \Sigma(0, 3, 6, 8, 13)$

